

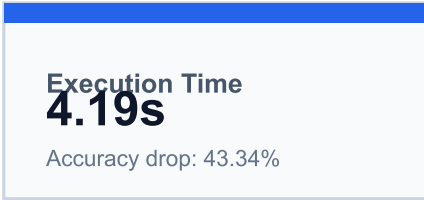
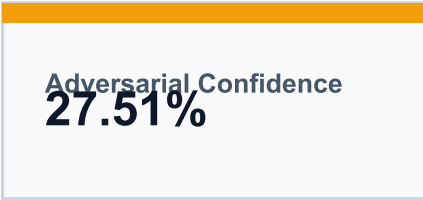
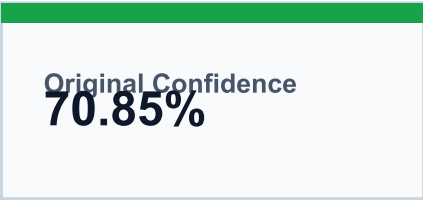
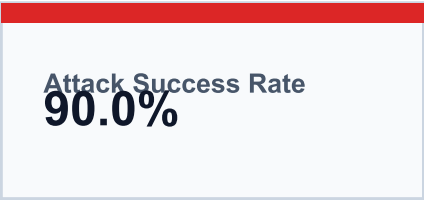
# Threat Assessment Report

Executive summary for model robustness under adversarial attack

## Model: microsoft/resnet-18

Attack Type: FGSM

Images Evaluated: 10



### Threat Level Assessment

# HIGH RISK

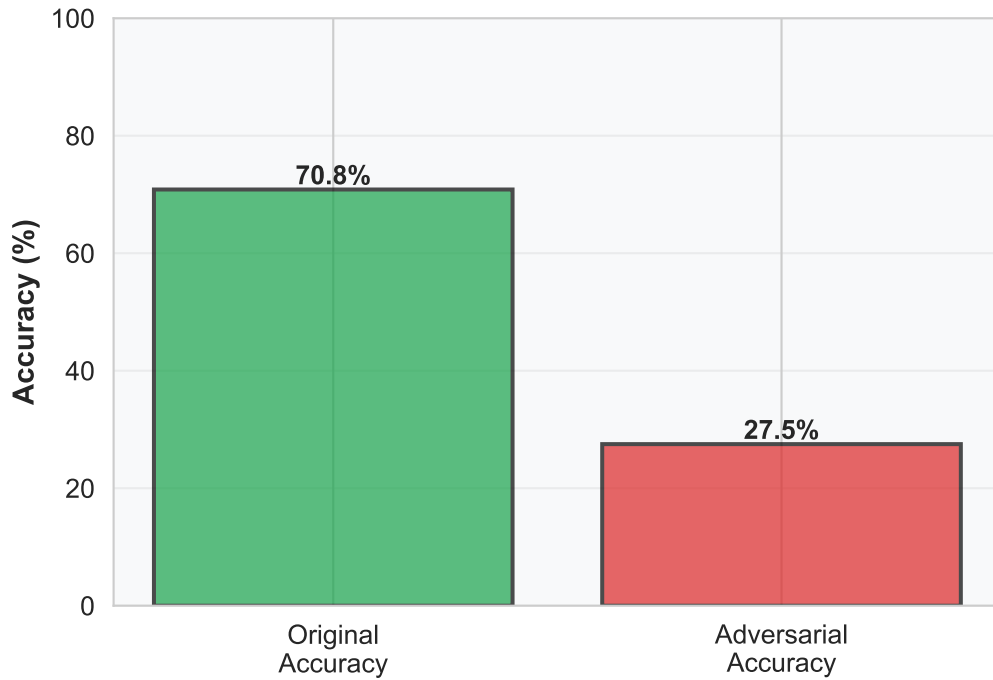
The FGSM attack altered model predictions on 90.0% of the evaluated images. Average confidence shifted from 70.85% to 27.51%, indicating the current robustness posture is high risk.

## Assessment Narrative

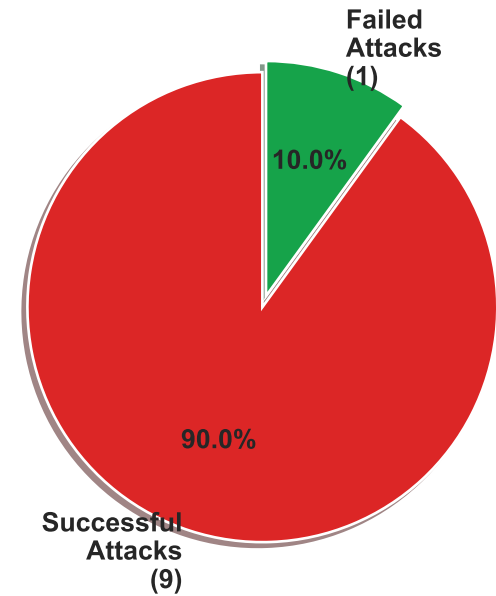
Successfully executed FGSM attack on model microsoft/resnet-18 using 10 test images. Attack success rate: 90.0%. Average original accuracy: 70.85%, Average adversarial accuracy: 27.51%. The attack successfully fooled the model in 9 out of 10 cases.

# Detailed Analysis

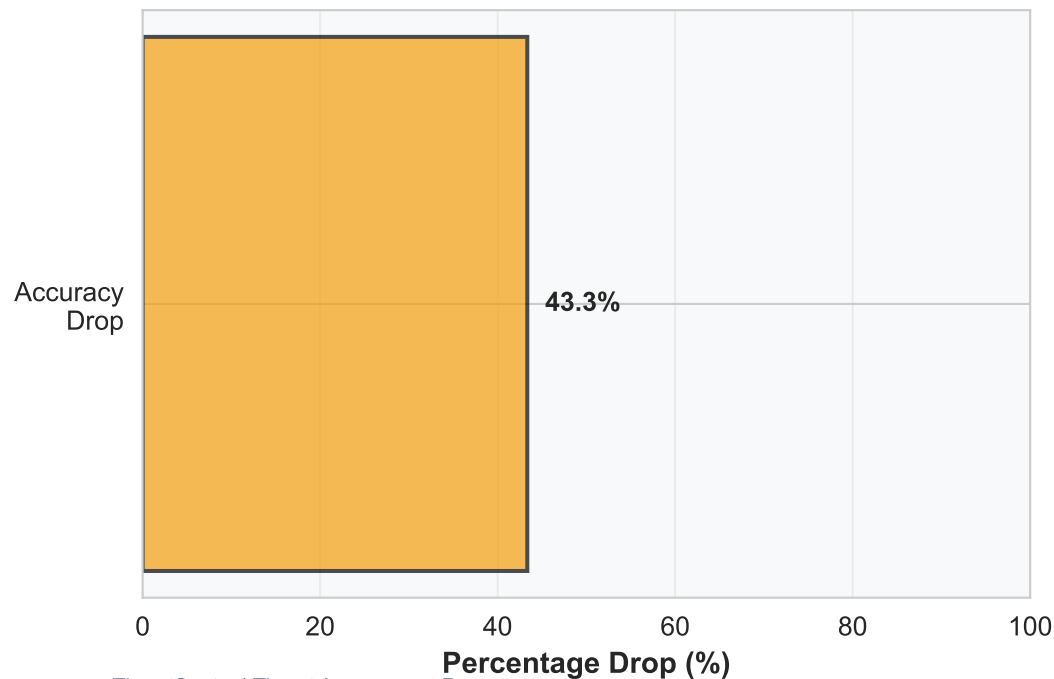
## Accuracy Comparison



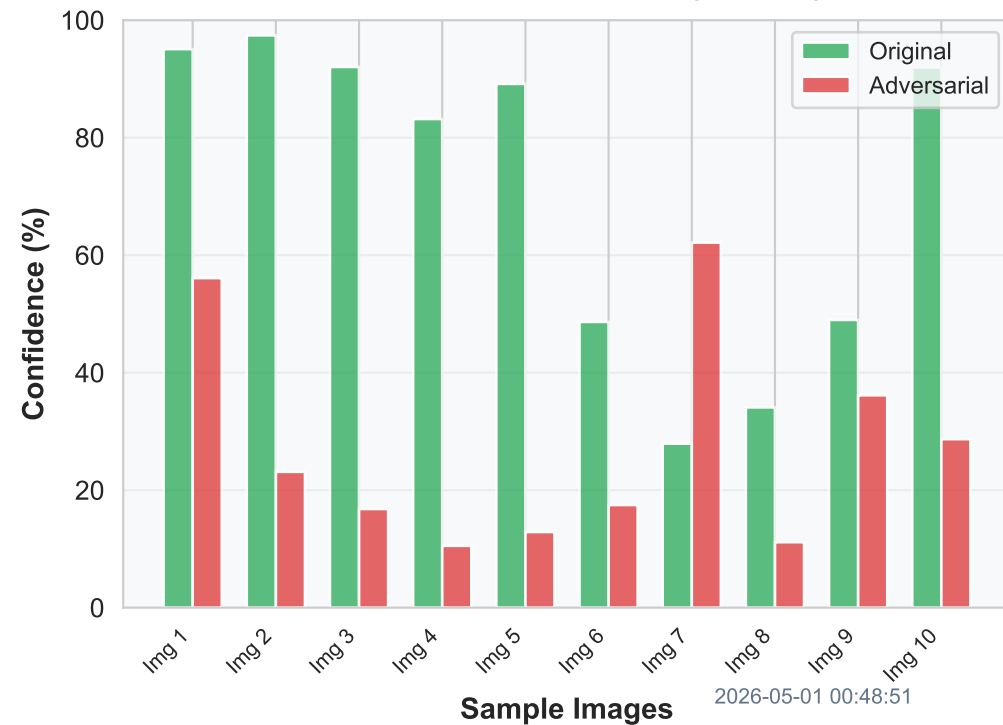
## Attack Success Distribution



## Model Robustness Impact



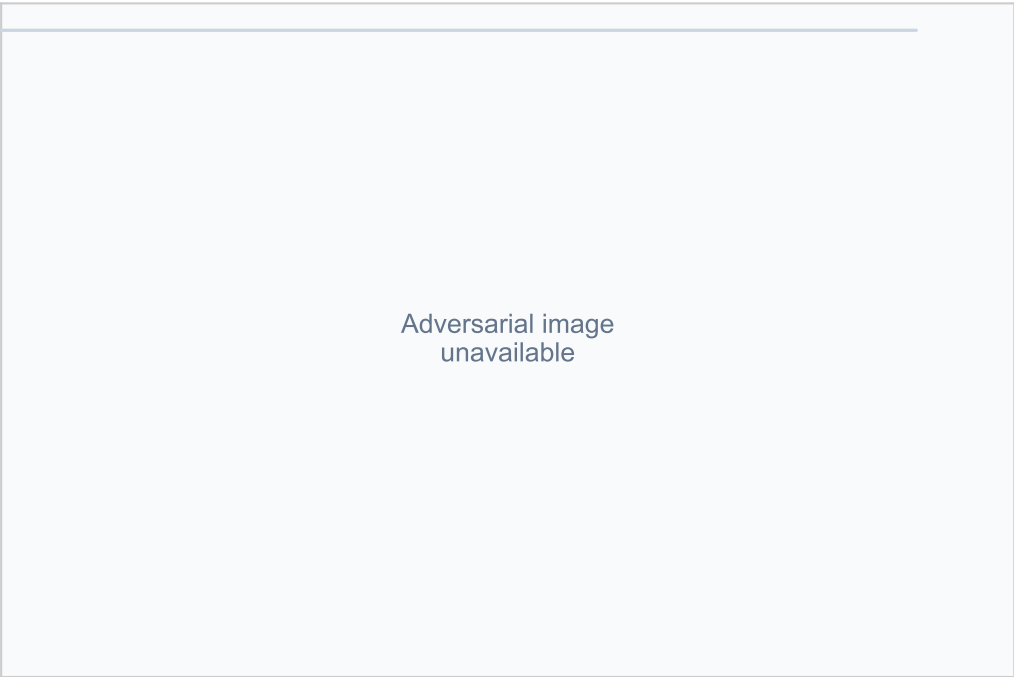
## Confidence Comparison (Sample)



# Image Evidence - Original vs Adversarial

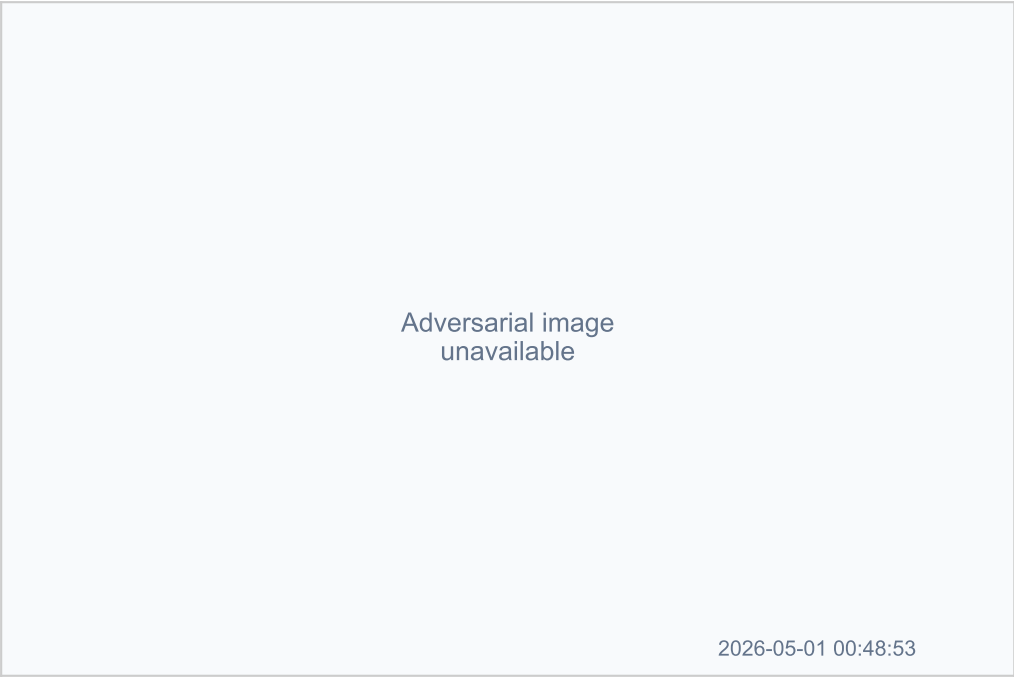
Images 10  
Original: hog, pig, grunter, squealer, Sus scrofa  
(27.9%)

Adversarial: water buffalo, water ox, Asiatic buffalo, Bubalus bubalis  
(62.1%)  
ATTACK SUCCESSFUL



Original: titi, titi monkey  
(89.2%)

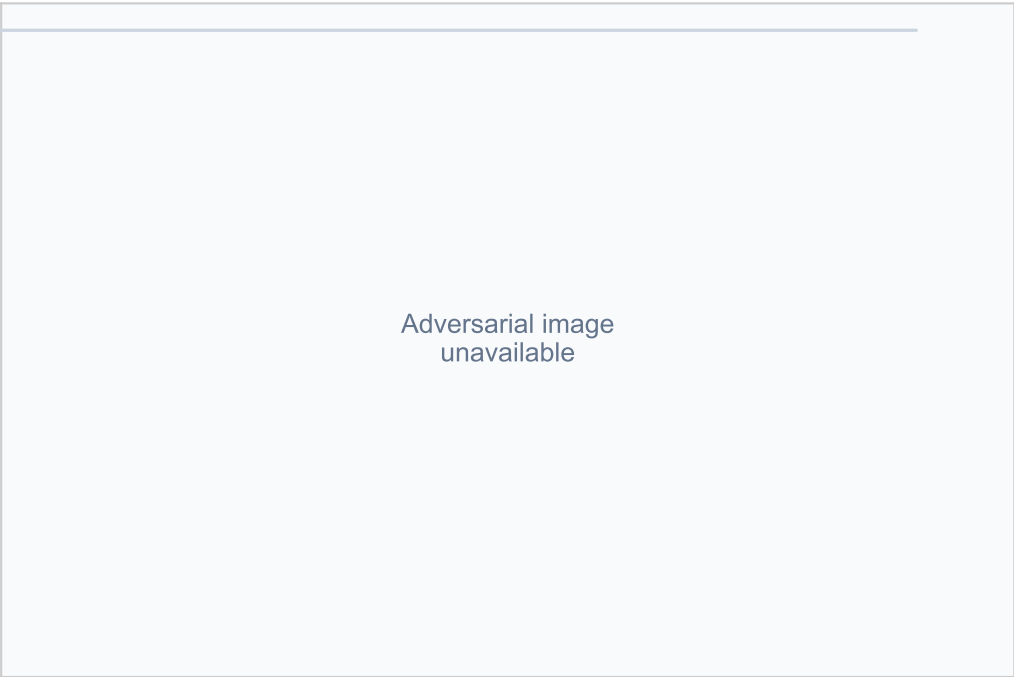
Adversarial: platypus, duckbill, duckbilled platypus, duck-billed platypus, Ornithorhynchus anatinus  
(12.9%)  
ATTACK SUCCESSFUL



# Image Evidence - Original vs Adversarial

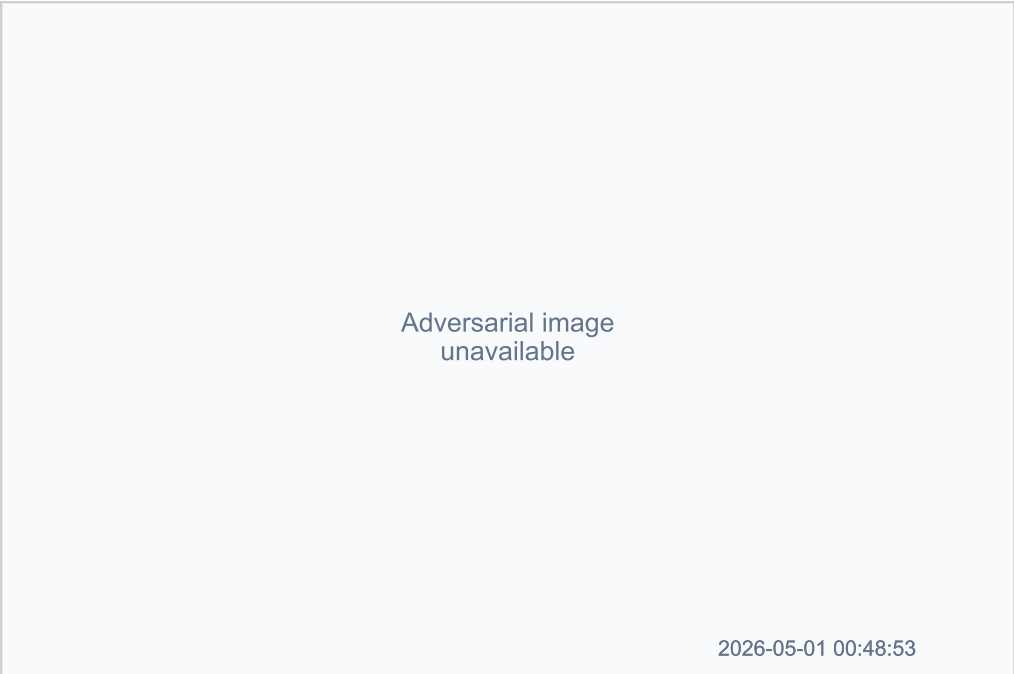
Images 10  
Original: hog, pig, grunter, squealer, Sus scrofa  
(27.9%)

Adversarial: water buffalo, water ox, Asiatic buffalo, Bubalus bubalis  
(62.1%)  
ATTACK SUCCESSFUL



Original: titi, titi monkey  
(89.2%)

Adversarial: platypus, duckbill, duckbilled platypus, duck-billed platypus, Ornithorhynchus anatinus  
(12.9%)  
ATTACK SUCCESSFUL

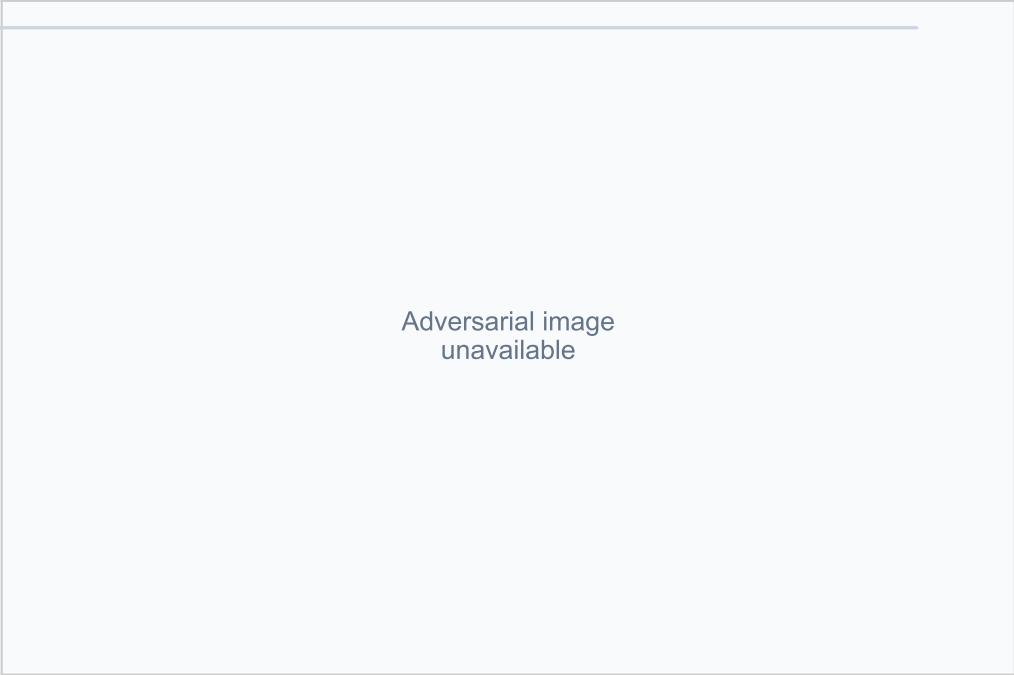




# Image Evidence - Original vs Adversarial

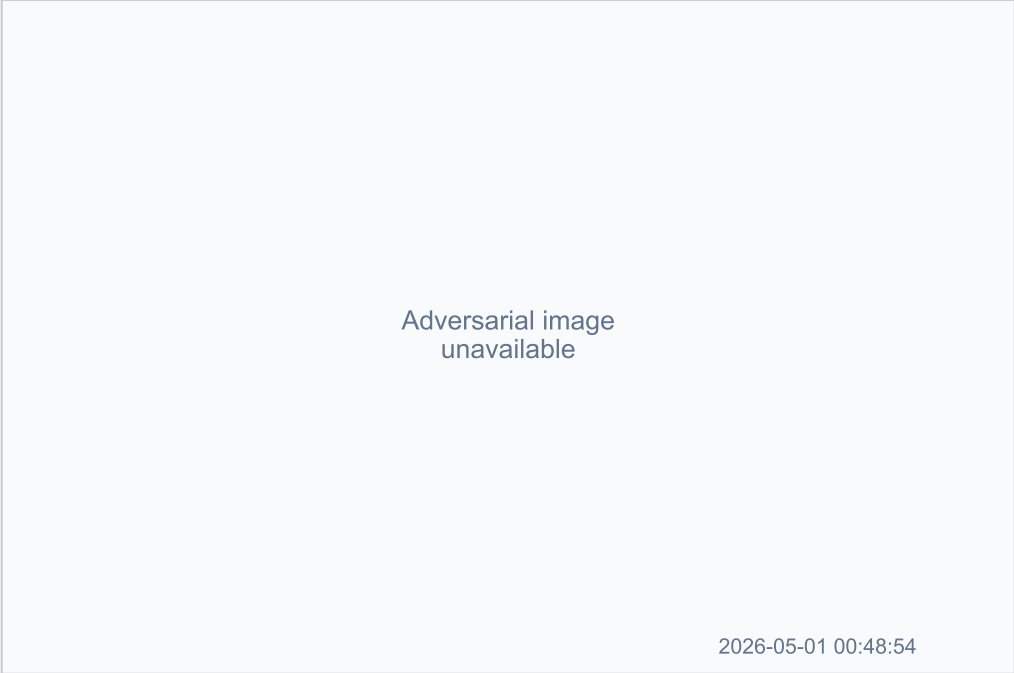
Original: water buffalo, water ox, Asiatic buffalo, Bubalus bubalis (97.4%)

Images 3-4 of 10



Adversarial: American alligator, Alligator mississippiensis (23.1%)  
**ATTACK SUCCESSFUL**

Original: coyote, prairie wolf, brush wolf, Canis latrans (91.9%)

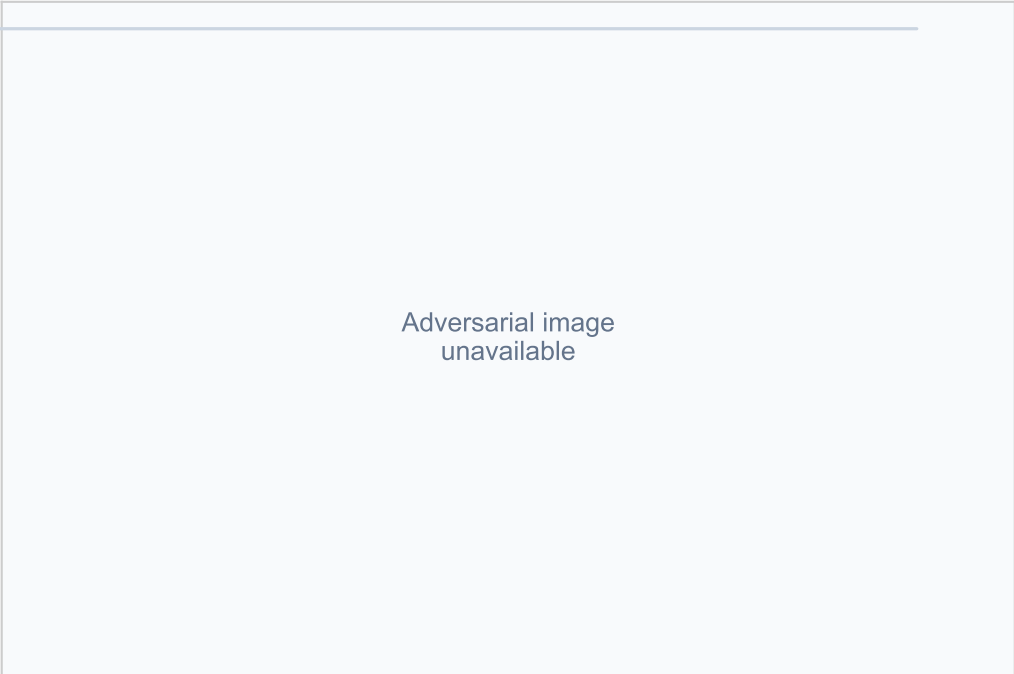


Adversarial: warthog (28.7%)  
**ATTACK SUCCESSFUL**

# Image Evidence - Original vs Adversarial

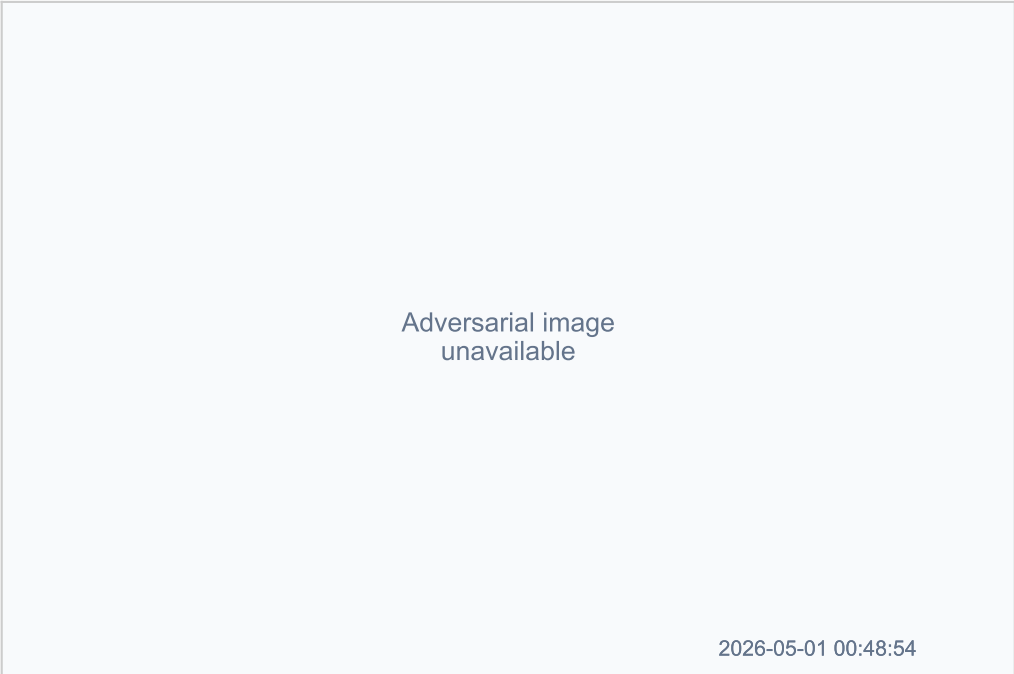
Original: water buffalo, water ox, Asiatic buffalo, Bubalus bubalis (97.4%)

Images 3-4 of 10



Adversarial: American alligator, Alligator mississippiensis (23.1%)  
**ATTACK SUCCESSFUL**

Original: coyote, prairie wolf, brush wolf, Canis latrans (91.9%)



Adversarial: warthog (28.7%)  
**ATTACK SUCCESSFUL**

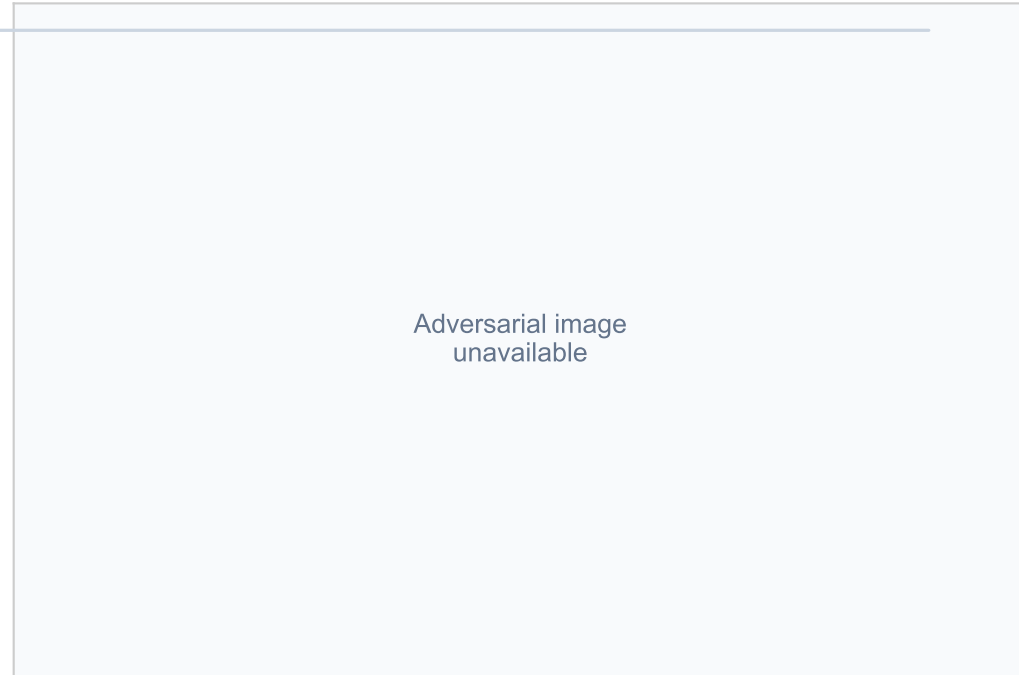


# Image Evidence - Original vs Adversarial

Original: sloth, *Hylobates syndactylus*, *Synphalangus syndactylus*  
(34.1%)

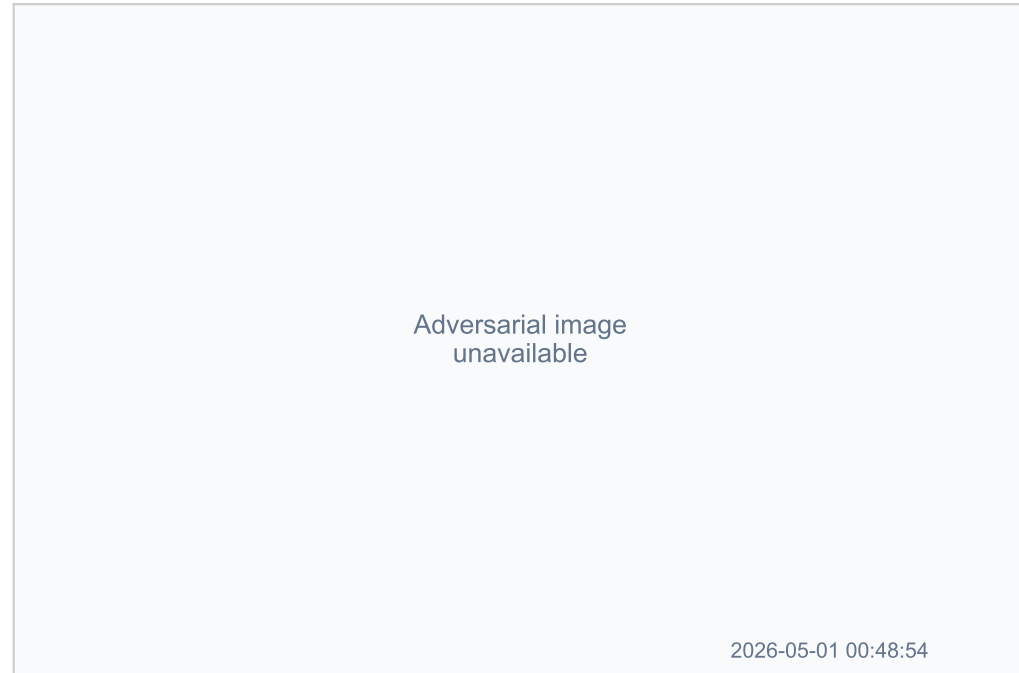
Images 5-6 of 10

Adversarial: bubble  
(11.2%)  
**ATTACK SUCCESSFUL**



Original: Indian elephant, *Elephas maximus*  
(49.0%)

Adversarial: Arabian camel, dromedary, *Camelus dromedarius*  
(36.2%)  
**ATTACK SUCCESSFUL**

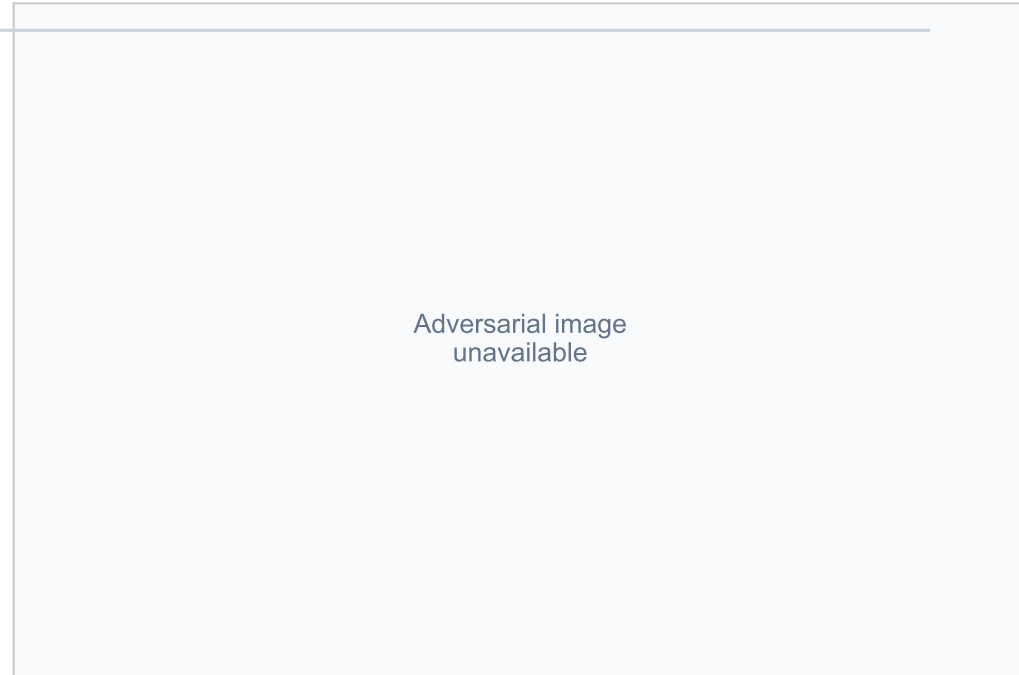


# Image Evidence - Original vs Adversarial

Original: Squirrel, *Hylocichla ustulata*, *Syrrhaptes ustulatus* (34.1%)

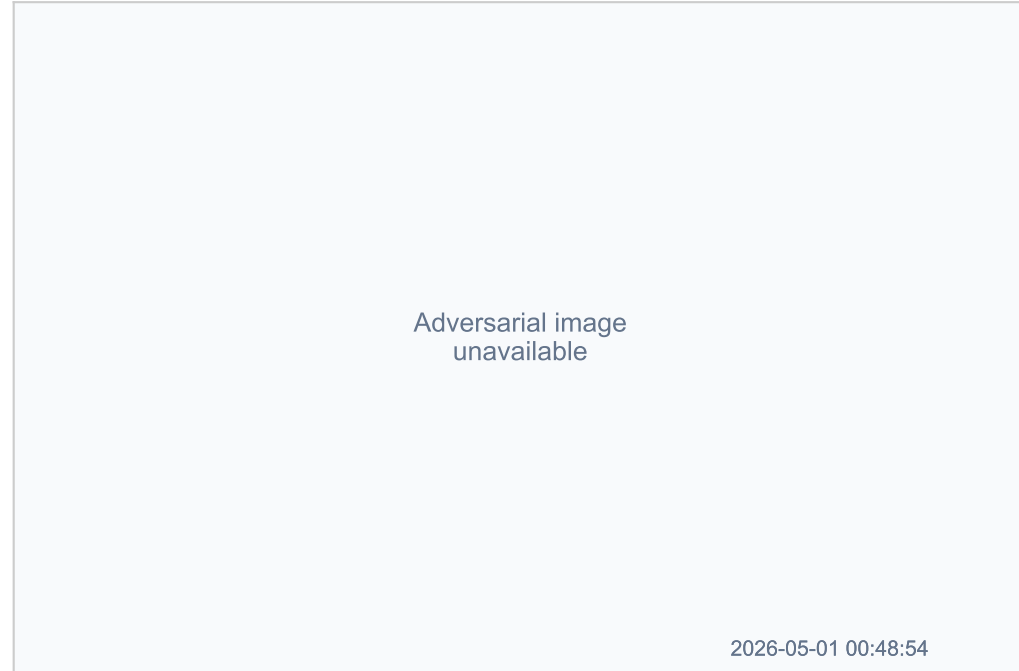
Images 5-6 of 10

Adversarial: bubble  
(11.2%)  
**ATTACK SUCCESSFUL**



Original: Indian elephant, *Elephas maximus* (49.0%)

Adversarial: Arabian camel, dromedary, *Camelus dromedarius* (36.2%)  
**ATTACK SUCCESSFUL**



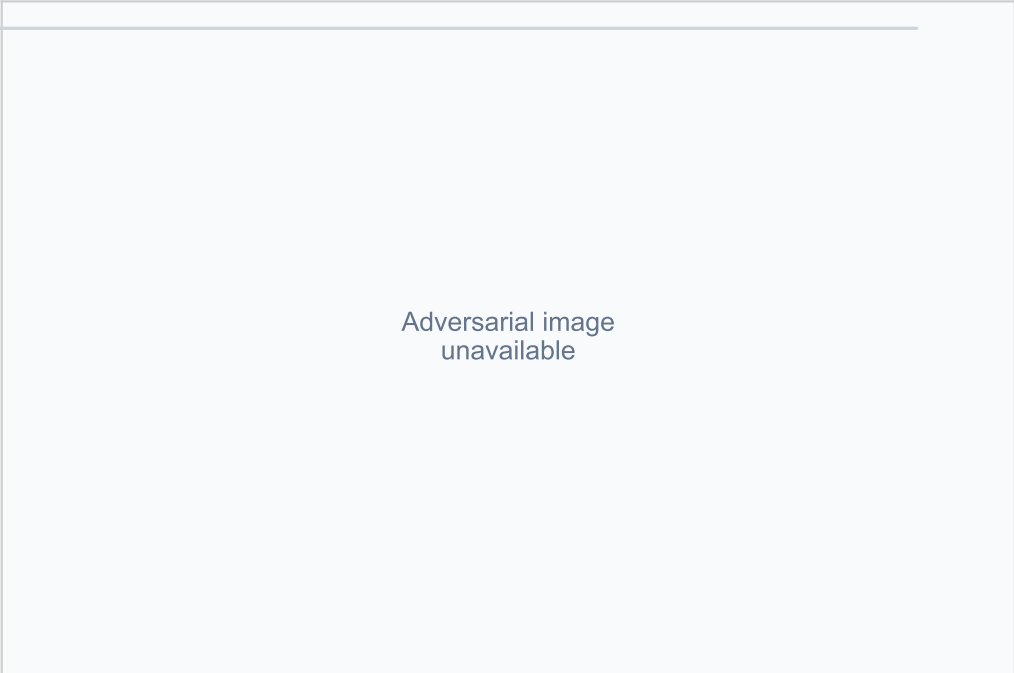
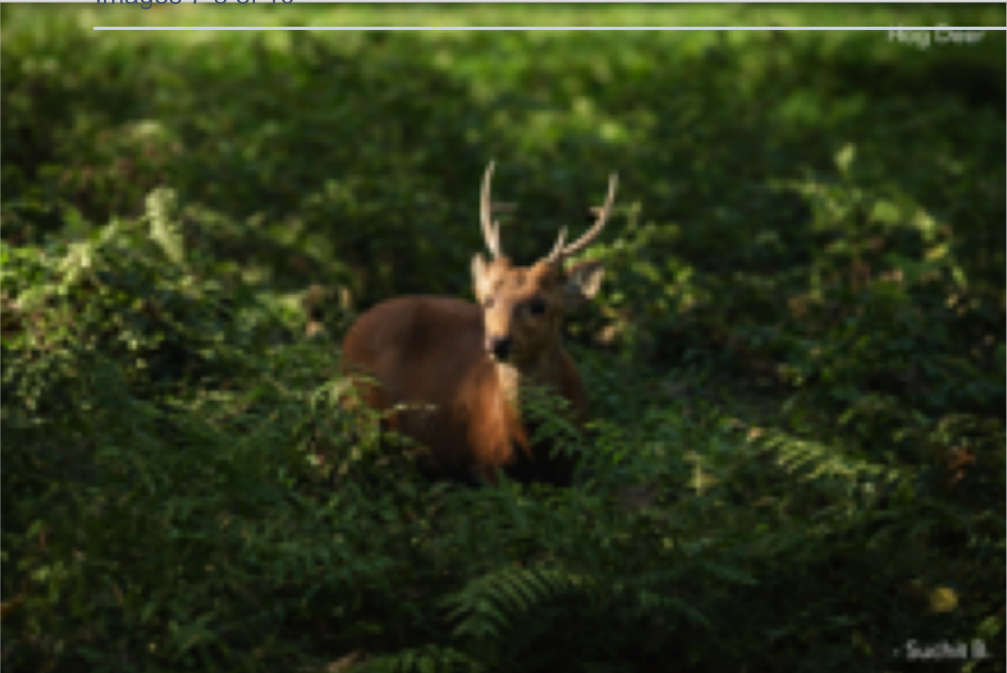


# Image Evidence - Original vs Adversarial

Images 7-8 of 10

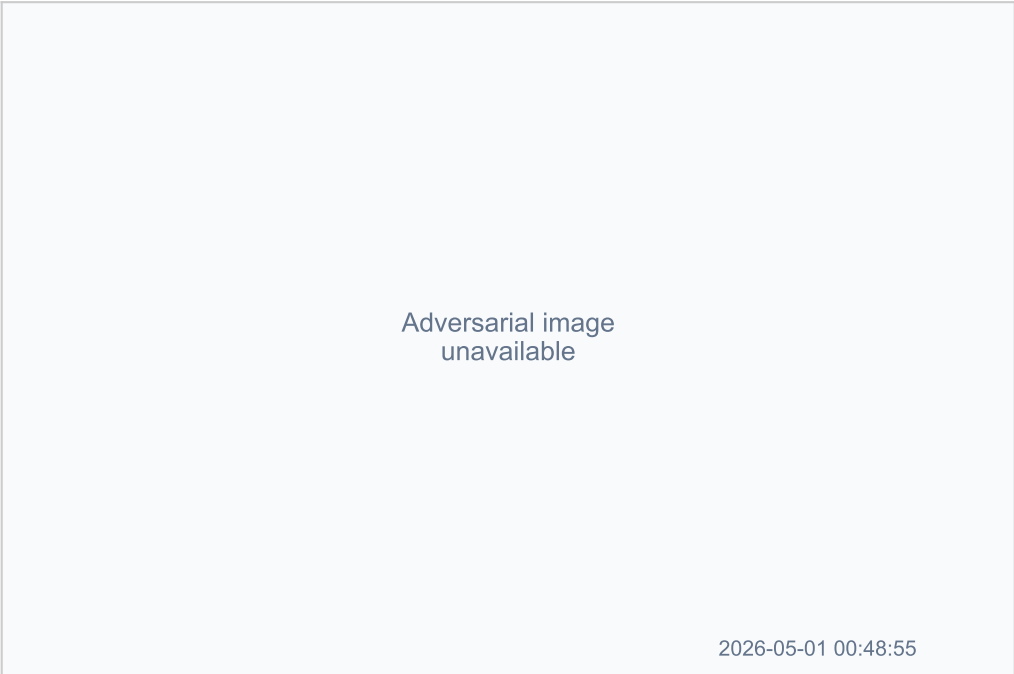
Original: fox, *Captiv. fox*  
(83.2%)

Adversarial: water snake  
(10.5%)  
ATTACK SUCCESSFUL



Original: dhole, *Cuon alpinus*  
(92.0%)

Adversarial: American alligator, *Alligator mississippiensis*  
(16.8%)  
ATTACK SUCCESSFUL

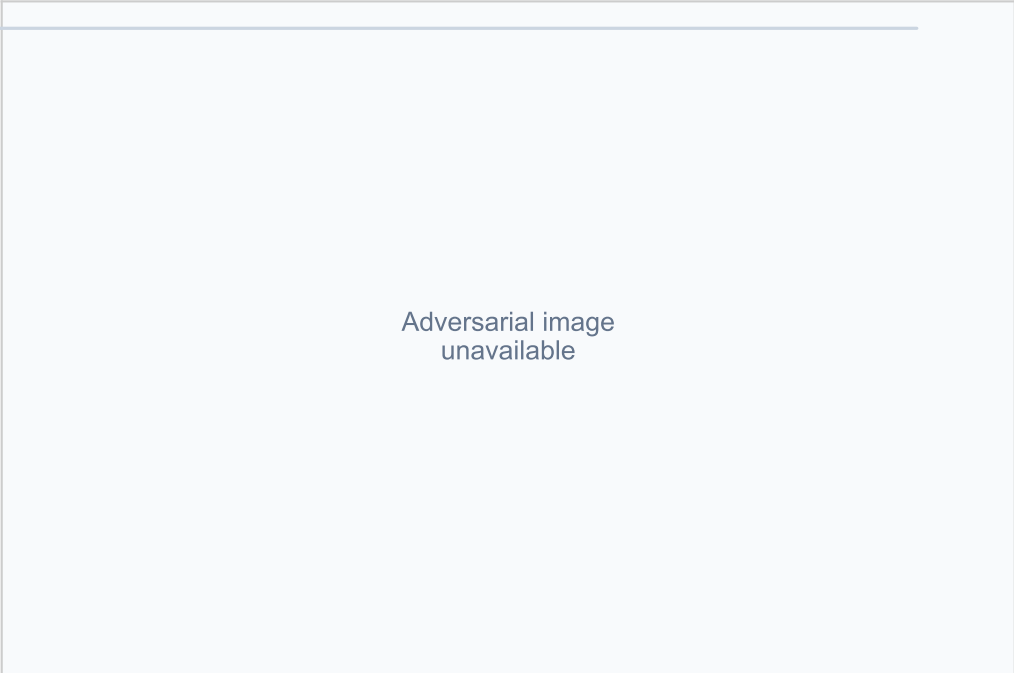
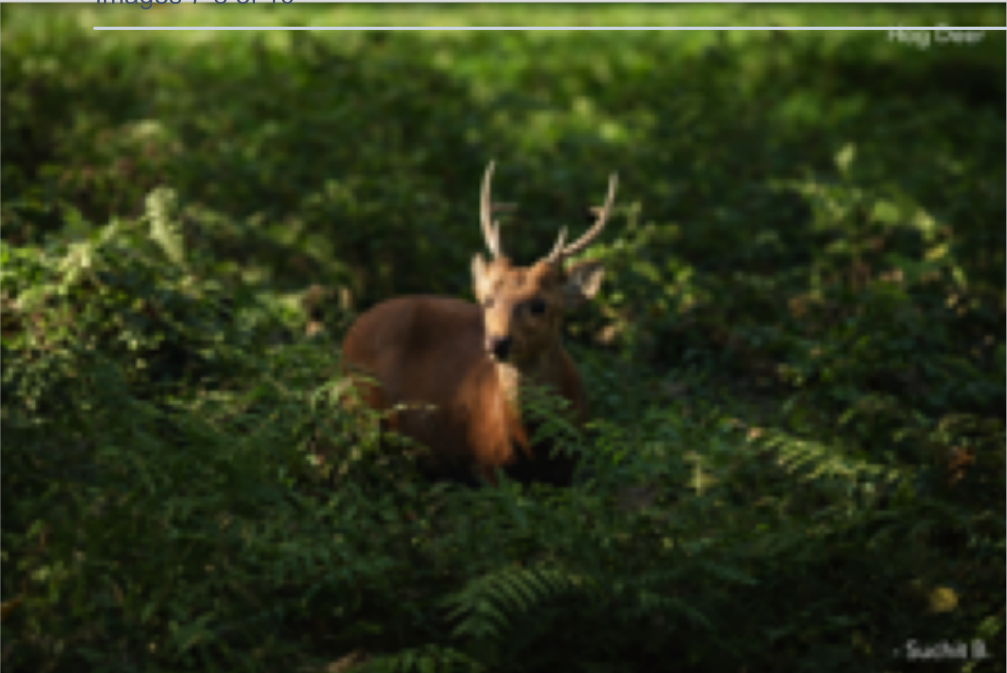


# Image Evidence - Original vs Adversarial

Images 7-8 of 10

Original: fox, *Captiv. fox*  
(83.2%)

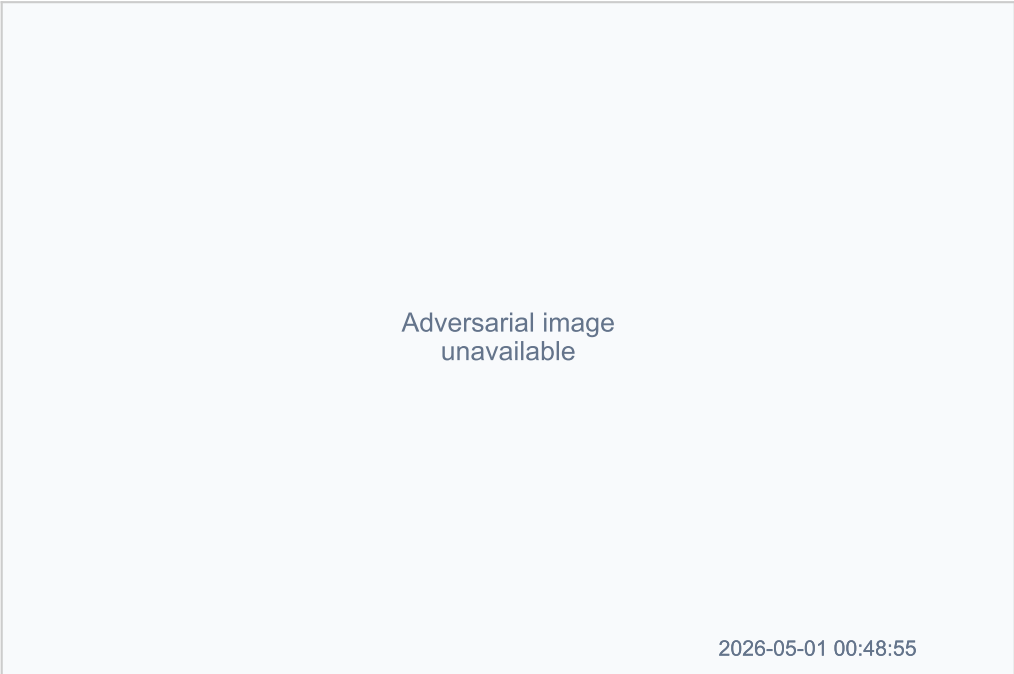
Adversarial: water snake  
(10.5%)  
ATTACK SUCCESSFUL



Adversarial image  
unavailable

Original: dhole, *Cuon alpinus*  
(92.0%)

Adversarial: American alligator, *Alligator mississippiensis*  
(16.8%)  
ATTACK SUCCESSFUL



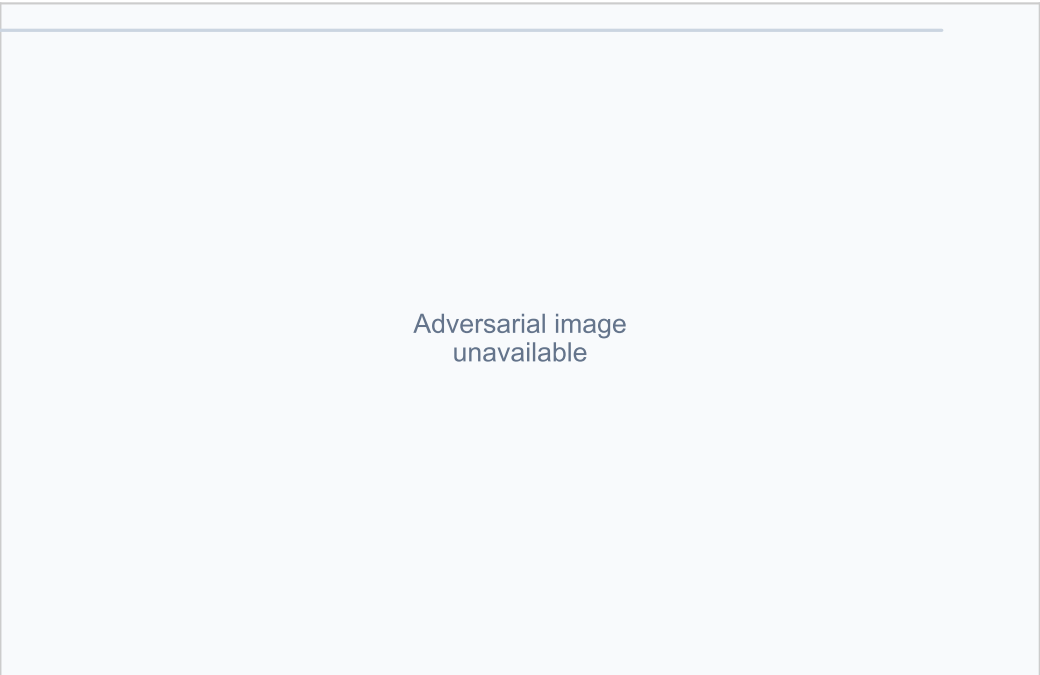
Adversarial image  
unavailable



# Image Evidence - Original vs Adversarial

Images 9-10 of 10

Adversarial: spider, web, spider's web  
(17.5%)  
**ATTACK SUCCESSFUL**



Adversarial image  
unavailable

Original: tiger, *Panthera tigris*  
(95.1%)

Adversarial: tiger, *Panthera tigris*  
(56.1%)  
**ATTACK BLOCKED**

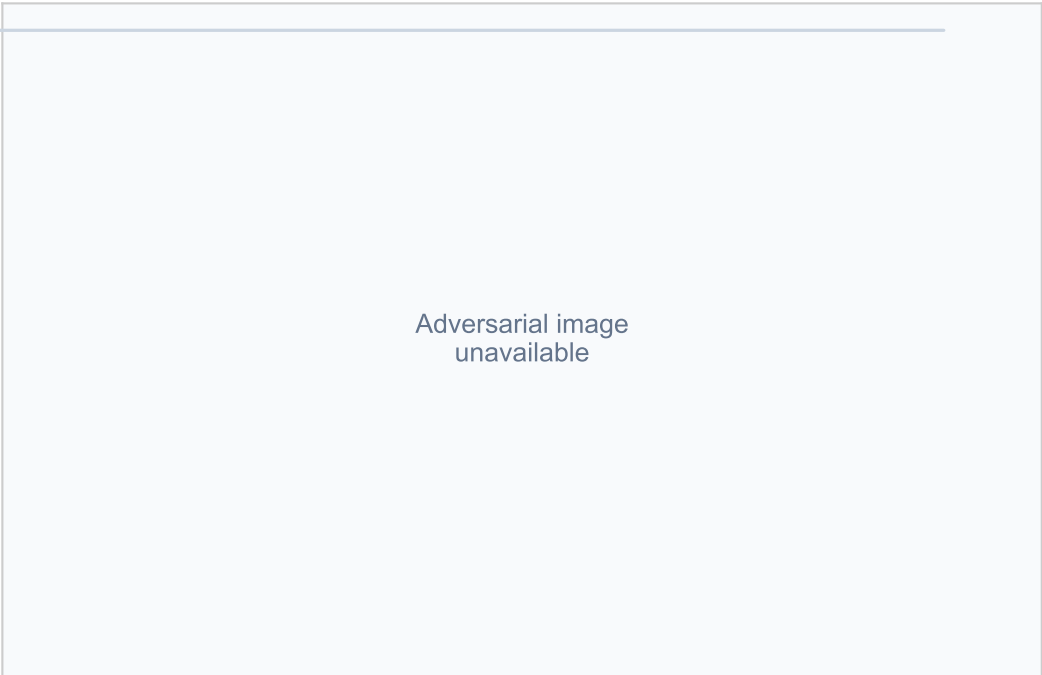




# Image Evidence - Original vs Adversarial

Images 9-10 of 10

Adversarial: spider, web, spider's web  
(17.5%)  
**ATTACK SUCCESSFUL**



Adversarial image  
unavailable

Original: tiger, *Panthera tigris*  
(95.1%)

Adversarial: tiger, *Panthera tigris*  
(56.1%)  
**ATTACK BLOCKED**



# Recommendations

Security actions based on observed threat exposure

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## Assessment Summary

Successfully executed FGSM attack on model microsoft/resnet-18 using 10 test images. Attack success rate: 90.0%. Average original accuracy: 70.85%, Average adversarial accuracy: 27.51%. The attack successfully fooled the model in 9 out of 10 cases.

## Security Recommendations

### 1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

### 2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

### 3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

### 4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

### 5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques